

JERUSALEM COLLEGE OF ENGINEERING
 (An Autonomous Institution Affiliated to Anna University, Chennai)
 Department of Science and Humanities
 Academic Year: 2024 - 25
QUESTION BANK
JPH2201 ENGINEERING PHYSICS II

*: K1 - REMEMBER, K2 - UNDERSTAND, K3 - APPLY, K4 - ANALYSE, K5 - EVALUATE, K6 -CREATE.

Unit 1 - SEMICONDUCTING MATERIALS

Course objective - To enlighten the students on the principles of semiconductor materials and electron transport properties

Course Outcome - Students will be able to be familiar with the functioning of semiconductors

PART - A

1	What is semiconductor? 1. The materials in which the electrical conductivity lies between conductors and insulators are called semiconductors. 2. They have resistivity value between 10^{-4} to $0.5 \text{ m}\Omega$. 3. The electrical conductivity of semiconductor increases, when we added to impurities and by increasing the temperature and it is contrary to the metals. 4. They have negative temperature Coefficient of resistance and which are formed by covalent bonds	CO1	K1
2	Write the properties of a semiconductor. 1. They are formed by covalent bonds. 2. They have small energy gap. 3. They have an empty conduction band and almost filled valence band 0K. 4. They have negative temperature coefficient of resistance. 5. They have resistivity of semiconductors lies between a semiconductor and Insulator. (10^{-4} to 0.5 Ohms m) 6. At 0 K it behave as insulator.	CO1	K1
3	Mention any Four advantages of semiconducting materials. 1. Semiconductors are smaller in size and use less power than other devices. 2. Semiconductors can be made to conduct electricity under certain conditions, such as heat, light, or voltage. 3. Impurities can be added to semiconductors to create new energy levels and alter their conductivity. 4. Semiconductors have a high melting point and low band gap energy, making them suitable for high-temperature applications.	CO1	K1

4	For an intrinsic semiconductor with a band gap of 0.7eV, determine intrinsic carrier concentration at T = 300K if $m_h^* = 6m_e^*$. Solution: Bandgap $E_g = 0.7 \text{ eV} = 0.7 \times 1.6 \times 10^{-19} \text{ V}$ $T = 300 \text{ K}$ Fermi energy for an intrinsic semiconductor $E_F = \frac{E_g}{2} + \frac{3KT}{4} \log \left[\frac{m_h^*}{m_e^*} \right]$ $E_F = \left[\frac{0.7 \times 1.6 \times 10^{-19}}{2} \right] + \left[\frac{3 \times 1.38 \times 10^{-23} \times 300}{4} \right] \log e^6$ $= 6.1563 \times 10^{-20} \text{ Joules}$ $E_F = \frac{6.1563 \times 10^{-20}}{1.6 \times 10^{-19}}$ Fermi energy level $E_F = 0.3847 \text{ eV}$	CO1	K3																											
5	<table border="1"> <thead> <tr> <th>S.No</th> <th>Elemental semiconductors</th> <th>Compound semiconductors</th> </tr> </thead> <tbody> <tr> <td>1.</td> <td>These are made from single element.</td> <td>These are made from compound (mixed) element.</td> </tr> <tr> <td>2.</td> <td>These are made from IV group and VI group elements</td> <td>These are made from III and V [or] II and VI elements.</td> </tr> <tr> <td>3.</td> <td>These are called as indirect band gap semiconductor (electron-hole recombination takes place through traps)</td> <td>These are called as direct band gap semiconductor (electron-hole recombination takes place directly)</td> </tr> <tr> <td>4.</td> <td>Heat is produced in the recombination recombination</td> <td>Photons are emitted during</td> </tr> <tr> <td>5.</td> <td>Life time of charge carriers is more due to indirect recombination</td> <td>Life time of charge carriers is less due to direct recombination.</td> </tr> <tr> <td>6.</td> <td>Current amplification is more</td> <td>Current amplification is less.</td> </tr> <tr> <td>7.</td> <td>These are used for making diodes, transistor, etc.</td> <td>These are used for making LED, laser diodes, etc.</td> </tr> <tr> <td>8.</td> <td>Example : Ge, Si</td> <td>Example : GaAs, GaP, CdS, MgO</td> </tr> </tbody> </table>	S.No	Elemental semiconductors	Compound semiconductors	1.	These are made from single element.	These are made from compound (mixed) element.	2.	These are made from IV group and VI group elements	These are made from III and V [or] II and VI elements.	3.	These are called as indirect band gap semiconductor (electron-hole recombination takes place through traps)	These are called as direct band gap semiconductor (electron-hole recombination takes place directly)	4.	Heat is produced in the recombination recombination	Photons are emitted during	5.	Life time of charge carriers is more due to indirect recombination	Life time of charge carriers is less due to direct recombination.	6.	Current amplification is more	Current amplification is less.	7.	These are used for making diodes, transistor, etc.	These are used for making LED, laser diodes, etc.	8.	Example : Ge, Si	Example : GaAs, GaP, CdS, MgO	CO1	K2
S.No	Elemental semiconductors	Compound semiconductors																												
1.	These are made from single element.	These are made from compound (mixed) element.																												
2.	These are made from IV group and VI group elements	These are made from III and V [or] II and VI elements.																												
3.	These are called as indirect band gap semiconductor (electron-hole recombination takes place through traps)	These are called as direct band gap semiconductor (electron-hole recombination takes place directly)																												
4.	Heat is produced in the recombination recombination	Photons are emitted during																												
5.	Life time of charge carriers is more due to indirect recombination	Life time of charge carriers is less due to direct recombination.																												
6.	Current amplification is more	Current amplification is less.																												
7.	These are used for making diodes, transistor, etc.	These are used for making LED, laser diodes, etc.																												
8.	Example : Ge, Si	Example : GaAs, GaP, CdS, MgO																												
6	Compare intrinsic and extrinsic semiconductor.(Write the previous answer)	CO1	K2																											

7	Write an expression for the concentration of electrons in the conduction band of an intrinsic semiconductor. $n = 2 \left(\frac{2 \pi m_e^* kT}{h^2} \right)^{3/2} e^{(E_F - E_C) / kT}$	CO1	K2
8	Write the expression for the concentration of holes in the valence band of an intrinsic semiconductor. $p = 2 \left(\frac{2 \pi m_h^* kT}{h^2} \right)^{3/2} e^{(E_v - E_F) / kT}$	CO1	K2
9	Write the expressions for carrier concentration in n-type and p-type semiconductor. $n = (2N_d)^{1/2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{3/4} e^{-\Delta E / 2kT}$ $p = (2N_a)^{1/2} \left(\frac{2\pi m_h^* kT}{h^2} \right)^{3/4} e^{-\Delta E / 2kT}$	CO1	K2
10	Define Fermi level in the case of semiconductors. Mention its position in intrinsic and extrinsic semiconductor at 0K. In a semiconductor, the Fermi level is the energy level that represents the highest occupied electron state at absolute zero temperature	CO1	K1

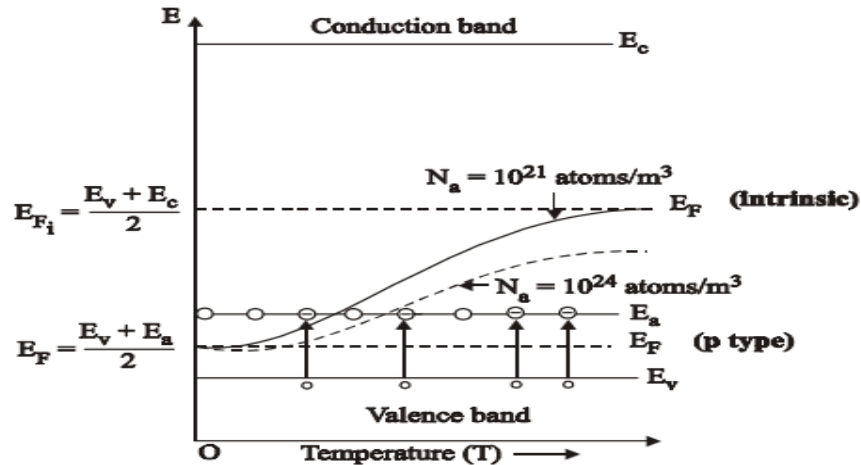


Fig. 2.12 Variation of Fermi level with temperature and impurity concentration in p-type semiconductor

11	Write an example for pentavalent and trivalent impurities. Pentavalent Impurities :Arsenic,Antimony,Bismuth Trivalent Impurities:Aluminium,Indium,Gallium	CO1	K1
12	Name any two compound semiconductors with applications. GaAs, GaP- These are used for making LED, laser diodes	CO1	K1
13	With increase of temperature the conductivity of a semiconductor increases. Why? With increase of temperature the conductivity of a semiconductor increases because they exhibit negative temperature coefficient of resistance.	CO1	K1
14	Define Hall-effect and Hall Voltage. When conductor (metal or semiconductor) carrying a current is placed in a transverse magnetic field, an electric field is produced inside the conductor in a direction normal to both the current and the magnetic field. This phenomenon is known as “Hall effect” and the generated voltage is called “Hall voltage”	CO1	K1
15	Mention the Uses of Hall effect. 1. The sign (N-type (or) P-type) of charge carriers can be determined. 2. The carrier concentration can be determined $\left[n = \frac{1.18}{qR_H} \right]$ 3. The mobility of charge carriers is measured directly $\left[\mu = \frac{-\sigma R_H}{1.18} \right]$. 4. Electrical conductivity can be determined $[\sigma = n q \mu]$. 5. It can be used to determine whether the given material is metal, insulator, or semiconductor and the type of the semiconductor. 6. It can be used to determine the power flow in an electromagnetic wave.	CO1	K1

PART B

1	Derive expressions for the density of electrons and holes in an intrinsic semiconductor. Hence deduce intrinsic carrier concentration.	CO1	K3
2	Obtain an expression for the density of electrons in conduction band in the case of N-type semiconductors.	CO1	K3
3	Obtain an expression for the density of holes in valence band in the case of P-type semiconductors.	CO1	K3
4	Explain the phenomenon of Hall effect. Derive an expression for Hall coefficient for n-type semiconductor and for p-type semiconductor.	CO1	K1

Unit 2 - MAGNETIC MATERIALS

Course objective - To acquaint them with the application of magnetic materials

Course Outcome - Students will be able to assess the magnetic properties of materials

PART - A

1.	Derive the relation between magnetic susceptibility and relative permeability. When a magnetic material is placed in a magnetic field (H), then two types of lines of induction passes through the material viz., • Due to magnetizing field(H) and • Due to the magnetized material(I) Hence total flux density , $B = \mu_o (H+I)$ But $\mu = B/H$ and $\mu_r = \mu / \mu_o$ Thus $\mu_o \mu_r H = \mu_o H(1+I/H)$ Since $\chi_m = I/H$ $\mu_r = \mu_o(1+ \chi_m)$	CO2	K2
2	What is Bohr magneton?	CO2	K1

	The orbital magnetic moment and the spin magnetic moment of an electron in an atom can be expressed in terms of atomic unit of magnetic moment called Bohr magneton. $1 \text{ Bohr Magnetron} = \frac{e\hbar}{2m} = \frac{eh}{4\pi m}$ $1 \Rightarrow \mu_B = 9.27 \times 10^{-24} \text{ Am}^2$		
3	Give- Curi-Weiss law and its importance. Curie - Weiss law is given by $\chi_c = \frac{C}{T - \theta}$ where $C \rightarrow$ Curie constant $T \rightarrow$ Absolute temperature $\theta \rightarrow$ Curie temperature Importance: It determines the susceptibility of the magnetic materials in terms of temperatures ie., If the temperature is less than curie temperature, a paramagnetic material becomes diamagnetic and if the temperature is greater than Curie Temperature, a ferromagnetic material becomes paramagnetic material.	CO2	K1
4	What is diamagnet? Mention any four properties of diamagnetic material. Diamagnet is a material which do not have permanent dipole moment. Properties • They repel magnetic lines of force • They acquire feeble magnetization in a direction opposite to applied field direction • Relative permeability is less than unity • Susceptibility is independent of temperature	CO2	K1
5	What is paramagnetism? Mention the properties of paramagnetic materials. Diamagnet is a material which have permanent dipole moment. Properties • They allow magnetic lines of force • They acquire magnetization in a direction of applied field direction • Relative permeability is greater than unity • Susceptibility is inversely proportional to temperature	CO2	K1

6	What is ferromagnetism? And mention its properties? Diamagnet is a material which have permanent dipole moment even in the absence of external magnetic field Properties • They allow more magnetic lines of force • They acquire high magnetization in a direction of applied field direction • Relative permeability is much greater than unity • Susceptibility is inversely proportional to temperature	CO2	K1
7	What is domain theory of ferromagnetism? The domain theory of ferromagnetism states that ferromagnetic materials are made up of small regions called domains, which are spontaneously magnetized. The domains align in the direction of an applied magnetic field	CO2	K1
8	Mention the energies involved in origin of domains in ferromagnetic materials. • Magneto static energy or Exchange energy • Crystalline Anisotropic energy • Bloch wall energy • Magnetostrictive energy	CO2	K1
9	What are ferrites? And mention its types. Ferrites are the materials in which the magnetic moments of unequal magnitude are aligned antiparallel. They are compounds of iron oxides with oxides of other metals. General formula is given by $X^{2+}Fe^{3+}O_4^{2-}$, where X^{2+} is divalent metal ions such as Fe^{2+}, Mg^{2+}, Zn^{2+} etc. There are two types 1. Regular spinel and 2. Inverse spinel	CO2	K1
10	What are the applications of ferrites? Ferrites are used in 1. Storage devices such as magnetic tapes, hard disc and in bubble memories 2. Producing ultrasonic waves 3. Radio receivers to increase sensitivity 4. Gyrator, isolator and circulator because of low eddy current and hysteresis loss	CO2	K1

11	Draw hysteresis loop and show the retentivity and coercivity in it.	CO2	K2															
12	What is hysteresis phenomenon? The lagging of magnetic induction (B) with respect to applied magnetic field intensity (H) is called hysteresis	CO2	K1															
13	What are soft and hard magnetic materials? (or) compare soft and hard magnetic materials on the basis of Hysteresis loop. Give examples. <table><tr><th>S. No</th><th>Soft magnetic materials</th><th>Hard magnetic materials</th></tr><tr><td>1.</td><td>Magnetic materials can be easily magnetised and demagnetised</td><td>Magnetic materials cannot be easily magnetised and demagnetised.</td></tr><tr><td>2.</td><td>They have high permeability</td><td>They have low permeability.</td></tr><tr><td>3.</td><td>Magnetic energy stored is not high</td><td>Magnetic energy stored is high</td></tr><tr><td>4.</td><td>Low hysteresis loss due to small hysteresis loop area</td><td>High hysteresis loss due to large hysteresis loop area.</td></tr></table>	S. No	Soft magnetic materials	Hard magnetic materials	1.	Magnetic materials can be easily magnetised and demagnetised	Magnetic materials cannot be easily magnetised and demagnetised.	2.	They have high permeability	They have low permeability.	3.	Magnetic energy stored is not high	Magnetic energy stored is high	4.	Low hysteresis loss due to small hysteresis loop area	High hysteresis loss due to large hysteresis loop area.	CO2	K2
S. No	Soft magnetic materials	Hard magnetic materials																
1.	Magnetic materials can be easily magnetised and demagnetised	Magnetic materials cannot be easily magnetised and demagnetised.																
2.	They have high permeability	They have low permeability.																
3.	Magnetic energy stored is not high	Magnetic energy stored is high																
4.	Low hysteresis loss due to small hysteresis loop area	High hysteresis loss due to large hysteresis loop area.																
14	Mention few magnetic storage devices. Hard disk drives (HDDs), floppy disks, and magnetic tape.	CO2	K1															

15	What do you understand by the term, “magnetic domain” and “domain walls”? A group of atomic dipoles organized into a tiny bounded regions called ferromagnetic domains. The boundary that separates two domains oriented in different directions is called domain wall or Bloch wall.	CO2	K2
----	---	-----	----

PART B

1	Briefly explain different types of magnetic materials and their properties and applications.	CO2	K2
2	Discuss Weiss theory(Domain theory) of ferromagnetism and explain the energies involved during magnetization.	CO2	K1
3	(i) What are Ferro magnetic domains? (ii) Draw a B-H curve for a ferro magnetic material and identify retentivity and coercive field on the curve.	CO2	K1
4	(i) Distinguish between soft and hard magnetic materials. (ii) Give an account of structure, properties and applications of ferrites.	CO2	K2

Unit 3 - SUPERCONDUCTING MATERIALS

Course objective - To impart knowledge on superconducting materials

Course Outcome - Students will be able to make use of superconductors in real life situations

PART - A

1.	What are superconductors? Mention its properties. Superconductors are the materials which conduct electricity almost without resistance. Properties • Resistivity is almost zero • They act as diamagnet • They act as superconductor below critical temperature and act as normal conductor above critical temperature • Critical temperature is directly proportional to pressure	CO3	K1
----	--	-----	----

2	Explain the phenomenon of super conductivity. The phenomena of losing resistivity absolutely when cooled to sufficiently low temperature is called super conductivity.	CO3	K2
3	What is transition temperature? The temperature at which a normal conductor loses its resistivity and becomes superconductor is known as critical temperature or transition temperature.	CO3	K1
4	Explain Meissner effect. When a super conductor is placed in a magnetic field, under the condition when $T \leq T_c$ and $H \leq H_c$, the flux lines are excluded from the material. Thus it acts as a diamagnet. This phenomenon is known as Meissner effect	CO3	K2
5	Explain the term critical magnetic field in superconductor. The minimum magnetic field required to destroy the superconducting property is known as critical magnetic field (H_c) It is given by the relation $H_C = H_o \left[1 - \left(\frac{T}{T_C} \right)^2 \right]$ $H_C \rightarrow$ Critical magnetic field at any temperature $H_o \rightarrow$ Critical magnetic field at absolute zero temperature $T_C \rightarrow$ Transition temperature of the material	CO3	K2
6	What are high T_c superconductors? Give examples. Any superconductor, if transition temperature is above 30K is called high T_c super conductor. Examples: $Y_1Ba_2Cu_3O_7$ $La_{1.85}Ba_{0.15}CuO_4$	CO3	K1
7	What are the applications of superconductors? It is used in magnetic levitation, cryotron and SQUIDS It is used in computer transmission lines It is used in imaging devices It is used for the production of high magnetic fields	CO3	K1

8	What is isotope effect in superconductivity? In a superconducting materials, transition temperature varies with the average isotopic mass M of its constituents. $\text{i.e., } T_C \propto \frac{1}{M^\alpha}$ $T_C = \frac{\text{constant}}{M^\alpha}$ $M^\alpha T_C = \text{constant}$ where α is called isotope effect coefficient and its value is 1/2.	CO3	K1
9	Define magnetic levitation? Super conductor behaves as a perfect diamagnet. Thus it repels magnetic lines of force. Thus when a magnet placed above a superconductor, it floats. This effect is magnetic levitation.	CO3	K1
10	What is SQUID? SQUID is the acronym for Superconducting Quantum Interference Device. It is a double junction quantum interferometer. Two Josephson junctions mounted on a superconducting ring forms this interferometer. It is an ultra-sensitive measuring instrument used for detection of very weak magnetic field in the order of 10^{-14} tesla.	CO3	K1
11	Super conducting tin has a critical temperature of 3.7K at zero magnetic fields and a critical field of 0.0306 Tesla at 0K. Find the critical field at 2K.	CO3	K3

	We know that $H_C = H_o \left[1 - \frac{T^2}{T_C^2} \right]$ Substituting the given values, we have $H_C = 0.0306 \left[1 - \frac{(2)^2}{(3.7)^2} \right]$ $H_C = 0.0306 \left[1 - \frac{4}{13.69} \right]$ $H_C = 0.0306 \left[\frac{13.69 - 4}{13.69} \right]$ $H_C = 0.0217 \text{ tesla}$		
--	--	--	--

PART - B

1	Explain the properties of Superconductors.	CO3	K2
2	Write an explanatory notes on BCS theory.	CO3	K1
3	Write a note on High Tc superconductors.	CO3	K1
4	Compare Type -I and Type II superconductor (or) Distinguish between soft and hard superconductors.	CO3	K2
5	Describe principle and working of Maglev train, cryotron and SQUID.	CO3	K1